

Thesis Progress

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1 Introduction

Robust biological systems maintain coherent global states despite local noise and damage, developing complex structures from initially homogeneous tissue through a process known as morphogenesis. Turing [7] was among the first to formalize how reaction-diffusion systems can spontaneously generate spatial patterns from uniform initial conditions, demonstrating mathematically that the interaction between two diffusing chemical species is sufficient to break spatial symmetry and produce stable, periodic structures. Building on this foundation, Gierer and Meinhardt [2] showed that pattern formation requires a short-range activator coupled with a long-range inhibitor, the activator promotes its own production locally while the inhibitor suppresses it at a distance, giving rise to stable concentration gradients that provide positional information to developing cells. Interestingly, we find the same behavior in neural activity as showed Amari [1] with its equations and the Mexican-hat kernel

Motivated by these dynamics, we extend the EngramNCA architecture [4] by introducing a slow ring-attractor controller (RA) into the GenePropCA network. EngramNCA separates morphogenetic growth into two components: a fast local update rule (GeneCA) that grows and maintains a target morphology (synaptic memory), and a gene propagation network (GenePropCA) that modulates gene expression across cells (RNA transfer). We augment GenePropCA with a structured low-dimensional recurrent controller whose state (a,b,d) evolves on a ring manifold and injects modulatory signals into the hidden representation of the fast network via a learned projection. This separation of timescales aims to improve robustness to morphological perturbations, especially rotations, a known challenge in NCA [5, 3], while providing an interpretable substrate for analyzing the emergent update rule.

1.1 Analytical study

Turing developed an equation able to describe the formations of pattern from reaction-diffusion in chemical substance

$$\frac{\partial x}{\partial t} = F(x) + D\Delta x$$

However, Turing's original formulation describe only the steady-state spatial patterns but not the oscillatory dynamics attractor that have been found (Gierer and Meinhardt, Amari) in different natural systems as a kind of persisten memory. To add this we will work with the continuos form of short-range activations and large-range inhibition of Amari:

$$\tau \frac{du}{dt} = -u + \int w(x - x') f(u(x')) dx' + I(x, t)$$

Where $w(x - x')$ represents the Mexican-hat kernel, which is the short-range activation with long-range inhibition, this convolution gives rise to lateral inhibition producing a stable attractor. We can linearize the equation by using a Taylor expansion around $x = x'$. Let $\epsilon = x' - x$ therefore

$$f(u(x)) = f(u(x + \epsilon))$$

By doing the Taylor expansion,

$$f(u(x')) = f(u(x)) + \epsilon f'(u(x)) + \epsilon^2/2 f''(u(x)) + \dots$$

Substituting in the integral

$$\int w(\epsilon) [f(u(x)) + \epsilon f'(u(x)) + \epsilon^2/2 f''(u(x)) + \dots] d\epsilon$$

We can split the integral into the kernel moments

$$f(u(x)) \int w(\epsilon) d\epsilon + f'(u(x)) \int \epsilon w(\epsilon) d\epsilon + f''(u(x))/2 \int \epsilon^2 w(\epsilon) d\epsilon$$

As the kernel is symmetric and centered at the origin

$$w(-\epsilon) = w(\epsilon)$$

Therefore, the odd moments vanish and the final equation is

$$\approx \hat{w}_0 f(u(x)) + \frac{\hat{w}_2}{2} f''(u(x)) + O(\epsilon^3)$$

In 1D we can approximate $f''(u(x)) = \Delta u$ which is the generalize form of the laplacian in 2D and as we are near of the center (stable ring, as we will see later) $f(u(x)) \approx u$

$$\approx \hat{w}_0 u + \frac{\hat{w}_2}{2} \Delta u$$

Therefore the Amari linearize equation is the follow one

$$\tau \dot{u} = (\hat{w}_0 - 1)u + K \Delta u + I$$

This linearize version of the Amari equation admits Hopf Bifurcation which is the case when 2 eigenvalues of the form $\lambda = \mu + iw$ in the imaginary axis cross

the origin. This lead us with unstable oscillation around the origin but with a stable ring with radius $r = \sqrt{\mu}$ which is the ring attractor geometry we want to implement on the network. The system can be describe with the following equations

$$\begin{aligned}\dot{r} &= \mu r - r^3 \\ \dot{\theta} &= w + br^2\end{aligned}$$

Where

- μ = stability at the origin
- w = frequency of the oscillations
- b = dependence of frequency with larger oscillations

The pattern formation occurs when $\mu > 0$ and a unstable spiral rise on the phase space with a stable circular at limit $r = \sqrt{\mu}$

We can linearize the equations by changing into cartesian coordinates

$$\begin{aligned}x &= r\cos\theta \\ y &= r\sin\theta\end{aligned}$$

$$\begin{aligned}\dot{x} &= \dot{r}\cos\theta - r\dot{\theta}\sin\theta \\ \dot{y} &= \dot{r}\sin\theta + r\dot{\theta}\cos\theta\end{aligned}$$

With $r^2 = x^2 + y^2$

$$\begin{aligned}\dot{x} &= (\mu r - r^3)\cos\theta - r(w + br^2)\sin\theta \\ \dot{y} &= (\mu r - r^3)\sin\theta + r(w + br^2)\cos\theta \\ \dot{x} &= \mu x - wy - (x^2 + y^2)x + b(x^2 + y^2)y \\ \dot{y} &= \mu y + wx - (x^2 + y^2)y + b(x^2 + y^2)x\end{aligned}$$

Whose at the same time gives origin to the Stuart-Landau equation by taking

$$\begin{aligned}A &= x + iy \\ \dot{A} &= \dot{x} + i\dot{y}\end{aligned}$$

Substituting $\dot{x}$ and $\dot{y}$

$$\dot{A} = (\mu x - wy) + i(wx + \mu y) - (x^2 + y^2)(x + iy) + b(x^2 + y^2)(-y + ix)$$

Identifying

$$(\mu x - wy) + i(wx + \mu y) = (\mu + iw)A$$

$$\begin{aligned}
-(x^2 + y^2)(x + iy) &= -|A|^2 A \\
b(x^2 + y^2)(-y + ix) &= b|A|^2 \cdot i(x + iy) = ib|A|^2 A
\end{aligned}$$

Therefore, the Stuart-Landau equation is

$$\dot{A} = (\mu + iw)A - c|A|^2 A$$

with $c = 1 - ib$

Moreover, when this equation is linearize near the limit circle $r = r^*$ a perturbation can be added

$$\begin{aligned}
r &= r^* + \rho \\
\rho &= r - r^* \rightarrow \dot{\rho} = \dot{r} \\
\dot{\rho} &= \mu(r^* + \rho) - (r^* + \rho)^3
\end{aligned}$$

As $\rho \ll 1$ and as $\dot{r} = \mu r^* - r^{*3} = 0$

$$\dot{\rho} = \mu\rho - 3r^{*2}\rho$$

As we work on the ring limit $r^* = \sqrt{\mu}$

$$\dot{\rho} = -2\mu\rho$$

When $\mu > 0$ we are on the stable ring. We can rewrite the Stuart-Landau equation in this regime: $|A|^2 \approx r^{*2} \approx \mu$

$$\dot{A} \approx (\mu + iw)A - \mu A = iwA$$

Which is only a rotating system but without damping oscillations that we just showed have the form $\dot{\rho} = -2\mu\rho$. Is for this that in the ring regime the Stuart-Landau equation become of the shape

$$\dot{A} \approx (-\alpha + iw)A$$

Where α capture the damping oscillations. Now we can linearize this system under the stable ring regime, which leads us to the following equations for $A = a + ib$

$$\begin{aligned}
\dot{a} &= -\alpha a + wb \\
\dot{b} &= -\alpha b - wa
\end{aligned}$$

The cubic terms disappear as we are linearizing around the stable ring (the attractor), therefore the dynamics will be governed by the radial velocity $(-\alpha)$. Putting all together, including the local spatial temporal results from Amari equations and considering it local for each cell $(c_{i,j})$, we found that

$$a_{i,j} \dot{=} -\alpha a_{i,j} + wb_{i,j} + K\Delta a_{i,j} + I^a$$

$$\dot{b}_{i,j} = -\alpha b_{i,j} - w a_{i,j} + K \Delta b_{i,j} + I^b$$

Following Turing’s framework, in which a slower-diffusing species is necessary to create spatial contrast and stabilize pattern formation, we introduce a diffusing leaky integrator d that acts as a slow spatial memory, smoothing the RA dynamics across neighboring cells.

$$\dot{d}_{i,j} = -\beta d_{i,j} + \kappa \Delta d_{i,j} + I^d$$

In these equations

- α : Radial damping
- w : Frequency of oscillation
- K : Spatial coupling strength
- β : Decay of injury/competence/adaptation
- κ : Spatial coupling of the diffusion
- Δ : 2D laplacian
- $I^{(a,b,d)}$: Local inputs derived from the cell environment such as the living mask, the edge magnitude, the Laplacian magnitude and the first two hidden layers of the cell channels as we will show on section 2.1.

Equations $\dot{a}$ and $\dot{b}$ will describe the dynamics of the RA that will be projected on a different time scale, every 4 steps, into the hidden states. Instead $\dot{d}$ equation will represent the lateral inhibition diffusion in the system, this will lead us understand to where the cells tends to activate (competence/injury). While d does not couple directly to (a,b) in the update equations, both contribute to the modulatory output that is projected onto the hidden NCA state, allowing the fast network to receive both oscillatory phase information from (a,b) and slow spatial context from d

2 Ring Attractor Controller Integration

We build upon the EngramNCA architecture, which consists of two coupled NCA networks: GeneNCA and GenePropNCA. The former acts as a synaptic memory in which simple growth primitives are learned; its learned weights are then used as latent information injected into GenePropNCA during training. In this way, the system learns dynamics encoded in a gene representation, functioning as an RNA-inspired memory transfer between the two networks.

It is well established that synaptic and RNA-mediated memory operate on different timescales: neural dynamics are faster than tissue-level growth. This motivates our decision to apply a slow Ring Attractor (RA) controller to GenePropNCA. The EngramNCA authors also acknowledge this distinction; however, in their experiments they were unable to find a measurable difference

when using two separate timescales for GeneNCA and GenePropNCA. To this end, we expand the channel vector by adding 6 channels: 3 dedicated to the RA controller (a,b,d) and 3 to the modulatory outputs that by this first approach they will consist on the ring attractor outputs. Our goal is to understand how the update rule dynamics evolve during training, and to evaluate whether the RA leads to a more robust system. Although NCAs are known to handle perturbations well, they still struggle with rotational invariance. We expect that the RA controller will improve robustness to spatial transformations, not only to rotational perturbations, but also to opening the door to translational movement across space (future work).

2.1 Addition of the RA

In addition to the new architecture configuration and the dynamic equations we established in Section 1.1, we need to establish which inputs will be used for the RA and how this RA will act on the network. For this we define a `slow_perception` function that will take:

- `alpha`/living mask of the cell
- The first two hidden channels of the state vector
- Local Edge
- Local laplacian of `alpha`

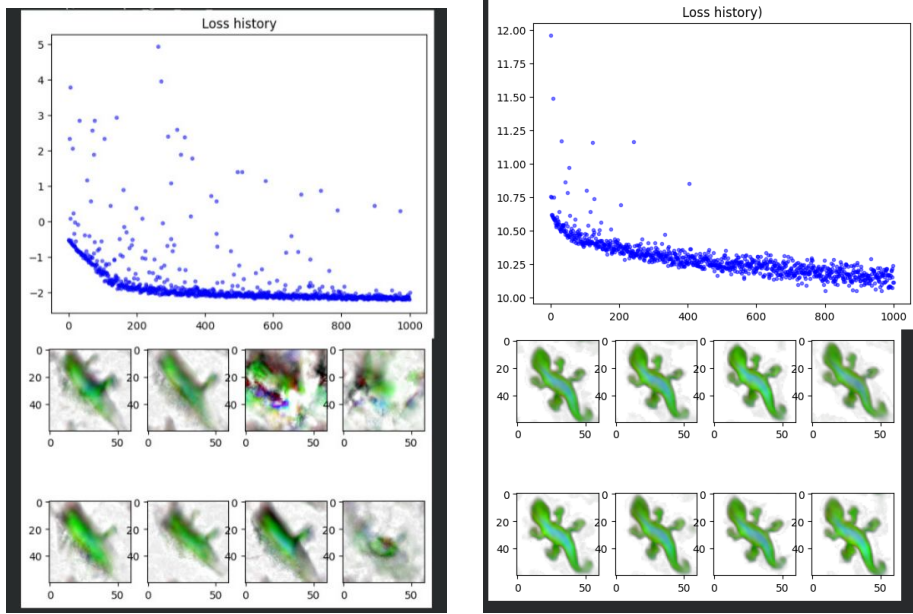
These will conform the input information that the RA will receive and will be update every 4 steps in comparison with the fast NCA. Then, the `discrete_update` will be apply with the equations obtained on Section 1.1 and a `consensus_update` will be done, we do this to simulate a Kuramoto-like synchronization which will consist on taking the local average of *a* and *b* between the neighbors of the cell.

After doing this the output will be stored on the modulation output that will be injected as a projection on the hidden channels by doing a convolution.

```
self.mod_proj = torch.nn.Conv2d(3, hidden_n, 1)
```

```
h = h + torch.tanh(self.mod_proj(mod))
```

It is important to note that the 6 extra channels are excluded from the standard perception vector. The RA channels do not represent observable cell state but instead act as an internal geometric controller, maintaining a spatial reference frame that guides the fast NCA dynamics. Their slower update rate, every 4 steps, reflects the biological distinction between fast neural signaling and slower tissue-level coordination: the RA must first observe how the system is evolving before applying a corrective modulation to the hidden state.



(a) First 1000 training steps with RA (b) First 1000 training steps without RA

Figure 1: Training progression of GenePropNCA with and without RA

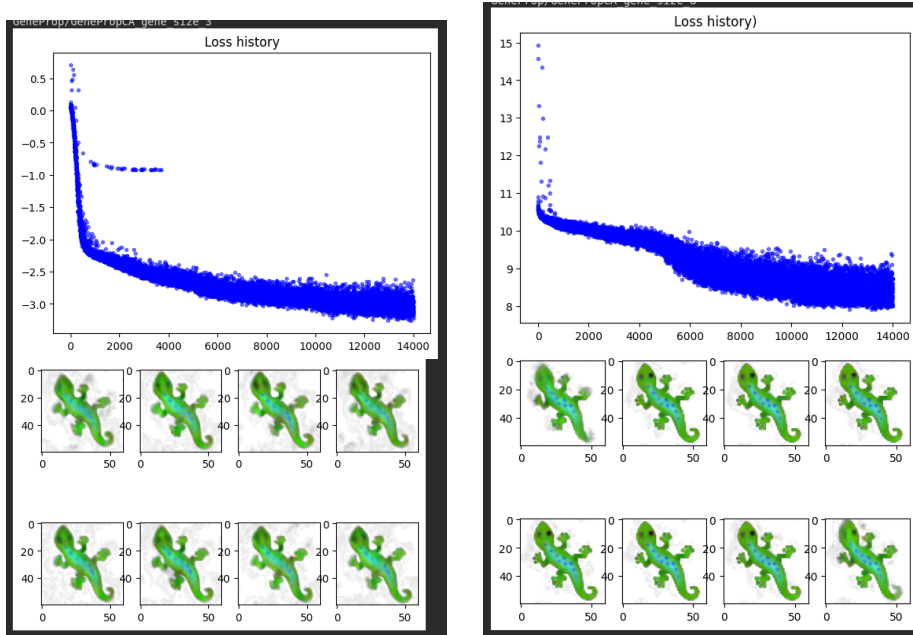
2.2 Results

After making these changes, we can study how the RA impact on the EngramNCA architecture. The first change that we can observe is that after training for the same time, the new architecture ended up more noisy with some cells active around the morphologic, this can be because now with the slow ring the exploration of the space around the lizard takes more time (additionally of the slow update)

2.3 Embryonic phase

During the growth of the lizard with the RA we can observe a more "embryonic" growth (Figure 1a) compared to EngramNCA (Figure 1b). The head and torso grow first, followed by the arms and finally the tail, while the standard architecture exhibits more homogeneous growth. Another difference observed during the training phase is that at an early stage, a more exploratory behavior is visible in the morphology. In Figure 1a we observe batches where an explosion of colors appears alongside more cells being active outside the lizard boundary.

At the end of training, after 14,000 iterations, we can compare the loss curves between both architectures. The EngramNCA produces a more detailed lizard but shows higher variance between batches, visible as a wider spread of points in the loss plot. In contrast, the RA architecture shows a narrower



(a) First 14000 training steps with RA (b) First 14000 training steps without RA

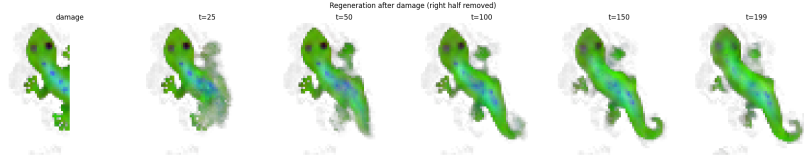
Figure 2: Training progression of GenePropNCA with and without RA

variance. We also observe that in the RA architecture, until approximately 5,000 steps, a subset of training samples appears stacked at a higher loss value before rejoining the general loss trajectory. This suggests that the RA, after an initial exploratory phase, learns to recognize not only the morphological shape but also the spatial distribution of cell colors.

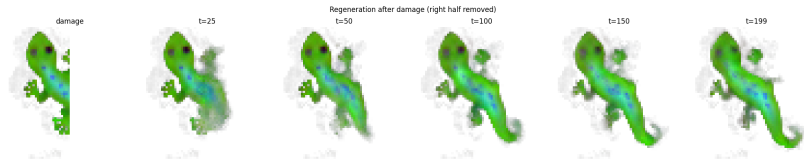
2.4 Damage

Both networks were trained only to grow the morphology and not to recover from damage. Nevertheless, we can examine whether each system is able to regenerate, maintain the morphology, or collapse after perturbation. To test this, we remove the right half of the image (a severe perturbation) and observe the system’s response.

In Figure 3 we observe the morphology after training. A first key observation is that regardless of the number of steps run, the system has learned to reproduce the image as a static pattern rather than growing the morphology dynamically, as we will observe with the RA. After damage, the morphology attempts to reconstruct itself but fails partially, an expected result given that the system was not trained to recover from damage. This is reflected in Figure 4, where the loss first decrease as some cells are trying to reconstruct the lizard but then increase as they were not able to recover the final shape



(a) Damage on EngramNCA after 10 steps of growing the morphology



(b) Damage on EngramNCA after 500 steps of growing the image

Figure 3: EngramNCA damage after 10 and 500 steps of growing the image

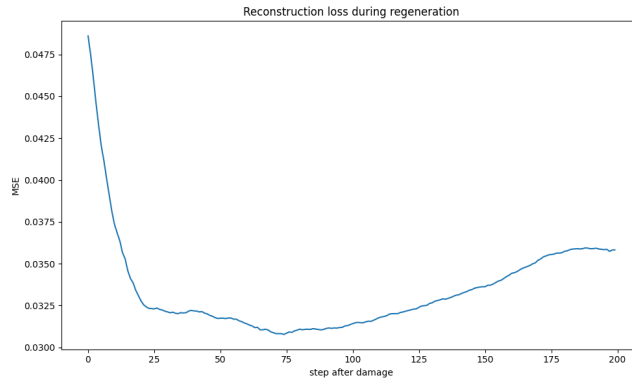
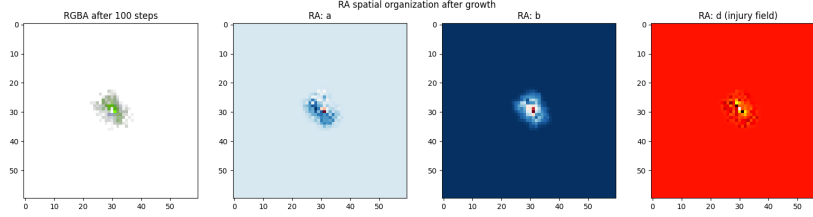
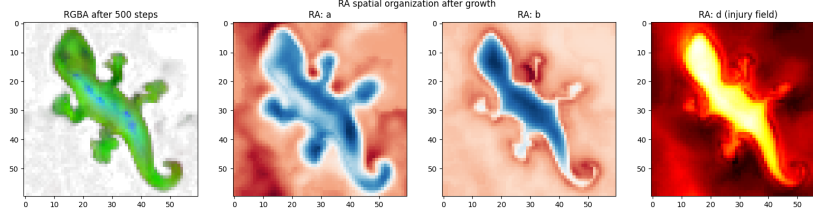


Figure 4: Loss of the system after the damage



(a) EngramNCA with RA after 10 steps of growing



(b) EngramNCA with RA after 500 steps of growing

Figure 5: Growing phase for the first 10 and 500 steps on the RA architecture

We repeat the same experiment for the new architecture with the RA. The first notable difference is that running the model for different numbers of steps produces visually distinct states, demonstrating that the morphology is actively growing rather than reproducing a static image as in EngramNCA. In Figure 5 we observe that after 10 steps a cluster of cells begins to grow, and after 500 steps the final morphology is reached, though with slightly less visual detail than EngramNCA. We also observe the spatial profiles of the (a, b, d) ring attractor state variables. The phase components a and b are spatially coordinated and define the boundary of the morphology, clearly separating it from the background. The field d , which acts as an injury, activation, or competence signal, is concentrated within the morphology, with higher activation at the center that gradually decreases toward the borders. This is a significant result, as it mirrors the behavior of morphogen gradients in developmental biology [6] where slow diffusion of chemical signals establishes positional information across a tissue. We conclude that the RA is learning the spatial structure of the morphology in a biologically interpretable manner.

After applying damage, we observe the following results. When damage is applied at an early growth stage (Figure 6a), the lizard is able to form a complete morphology even though the system was not trained for damage recovery. In contrast, when damage is applied after the lizard is fully formed (Figure 6b), the system struggles to reconstruct the original shape.

A further result is obtained by plotting the RA amplitude and reconstruction loss after damage (Figure 7). In the early-stage case, the loss starts high but rapidly decreases and stabilizes, indicating successful recovery. In the fully-

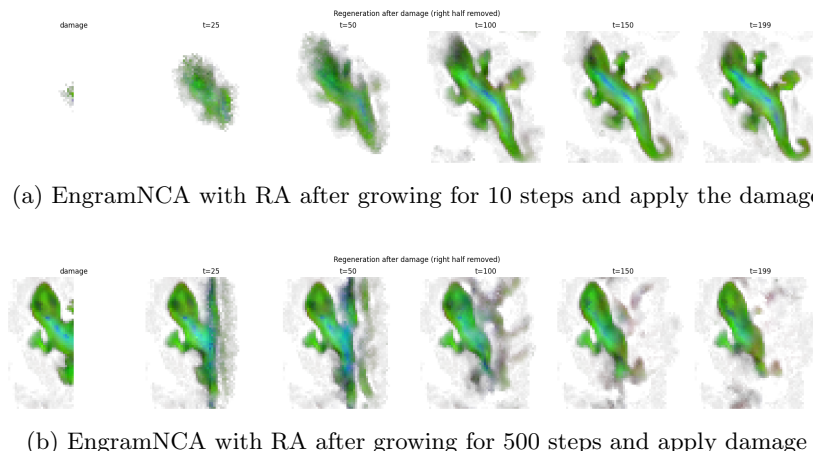
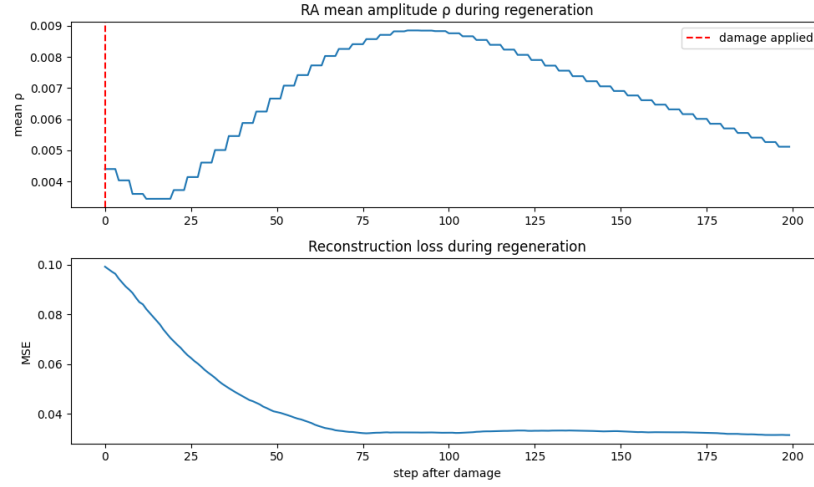


Figure 6: Damage applied after growing for the first 10 and 500 steps on the RA architecture

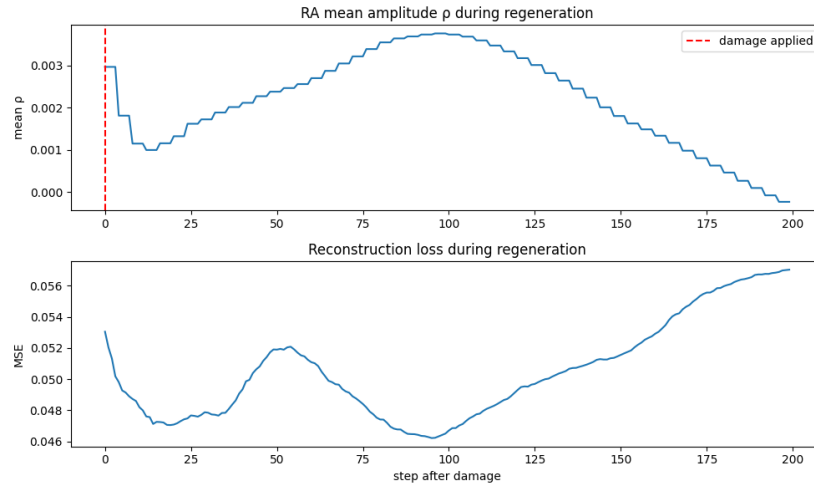
formed case, the loss initially decreases slightly as the system attempts reconstruction, but subsequently increases as recovery fails. In both cases, the RA amplitude shows the same behavior: it drops immediately after damage then rises, oscillating as the controller reorganizes the spatial state of the cells. We interpret this as evidence that the slow controller is actively responding to the perturbation by reorganizing cell competence, even in the absence of explicit damage training.

An additional analysis examined the correlation between the alpha channel and the injury scalar field d . The Pearson correlation coefficient between these two quantities is $r = 0.844$, indicating a strong positive relationship, even if they are intrinsically correlated since the living mask is an input. This result suggests that the competence field (d) is spatially coupled to cell density; regions where cells are alive and active (high alpha) correspond to regions of high competence signal. This is consistent with the morphogen gradient interpretation: the slow controller concentrates its activation signal where the morphology is present, providing a spatial reference that could guide regenerative behavior under damage.

The last analysis consists of the evolution of the equation parameters ($\alpha, \omega, K, \kappa, \beta$) during training. We do this to determine whether the system reaches a stable attractor after the training. In Figure 8 we observe that α remains positive throughout training and converges to $\alpha \approx 1.0$, confirming that the controller operates within the stable ring attractor regime established by the theoretical framework. The frequency ω initially increases alongside α but begins decreasing around iterations 500–1000, eventually reaching zero. This indicates that the network transitioned from an exploratory rotational regime, in which the phase actively searches the ring, to a static phase configuration consistent with the static morphogenetic target. This transition is consistent with the train-



(a) Amplitude mean the MSE loss after applying damage on the first 10 steps



(b) Amplitude mean and MSE loss after applying damage on the first 500 steps

Figure 7: Mean RA amplitude of the systems and MSE after applying damage for 10 and 500 growing steps

ing curve in Figure 2a, where a subset of batches stalls at a higher loss before rejoining the main trajectory. In future implementations targeting a moving morphology, we expect ω to remain nonzero as the phase continuously tracks spatial displacement

In the diffusion and coupling plot, K reaches a peak around iteration 2000 before decaying to $K \approx -0.4$. Negative K means the Laplacian acts as anti-diffusion on the (a, b) channels where rather than smoothing spatial gradients, it sharpens boundaries, which explains the morphological edge visible in Figure 5. The system remains stable despite negative K because the radial damping $-\alpha a$ bounds amplitude growth; the interplay between anti-diffusion and damping places the controller near the boundary of linear stability, with nonlinear dynamics preventing divergence. On the other hand, both β and κ converge to positive steady-state values with $\kappa < \beta$, placing the d field in a decay-dominated regime in which the competence signal is spatially concentrated rather than broadly diffused, this is consistent with the high activation core observed in Figure 5.

The implied ring radius $r^* \approx 1.0$ stabilizes after approximately 3000 iterations and remains constant for the remainder of training. The convergence of all parameters to geometrically consistent values, in particular $\alpha > 0$ without explicit constraint, confirms that the morphogenetic objective is compatible with ring attractor dynamics and that the system reaches a stable attractor through training alone

3 Conclusions and Future studies

We partially implemented the ring attractor controller by introducing the (a, b, d) state variables and the modulatory projection into the hidden space of GenePropCA. Although the FiLM conditioning mechanism was not yet implemented, the system successfully learned to grow the target morphology, demonstrating that the ring attractor dynamics do not destabilize training, one of the primary risks identified in the project plan. Beyond basic morphogenesis, the system exhibits several emergent properties that support the central thesis claim. First, the model grows the morphology dynamically from a single-pixel seed, rather than reproducing a static pattern, indicating that the slow controller provides a meaningful spatial reference during growth. Second, the strong correlation ($r=0.844$) between the injury field d and the alpha channel confirms that the RA has learned to couple its competence signal to cell density, producing a spatial organization consistent with morphogen gradient behavior in developmental biology. Third, the ring attractor state variables (a, b) develop spatially coherent patterns that follow the morphological boundary, and the amplitude responds measurably to damage even without explicit perturbation training. Fourth, the learned parameters of the dynamic equations converge to theoretically consistent values without explicit constraints: α remains positive throughout training, confirming that the network independently discovered the ring stability condition, and ω converges to zero, reflecting the static nature of the morphogenetic target.

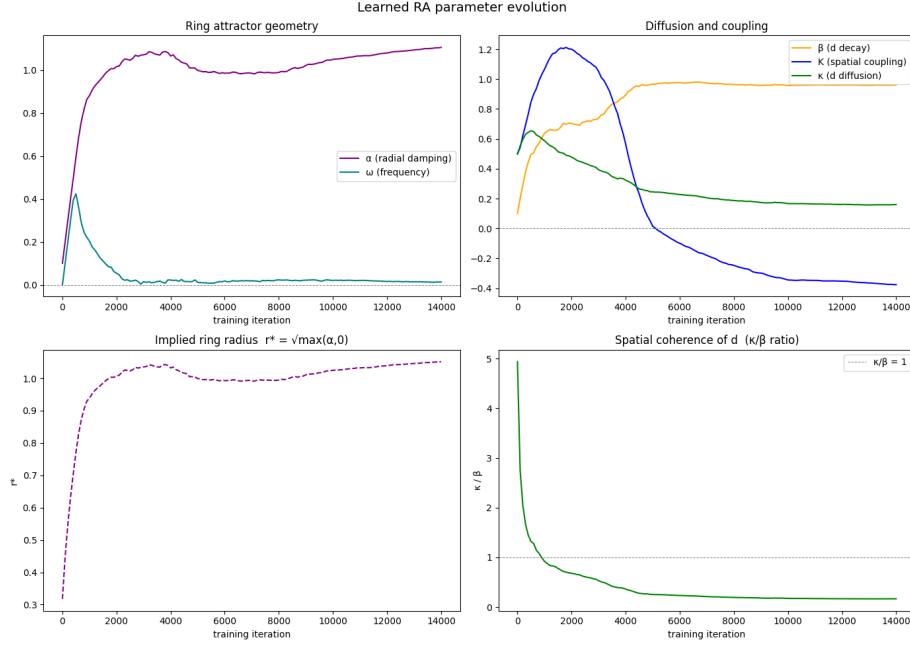


Figure 8: Evolution of the parameters during training

This constitutes independent empirical validation of the theoretical framework derived in Section 1.1. Together, these results suggest that local interactions governed by the slow controller give rise to a globally coherent steady state, a key property of biological morphogenetic systems.

3.1 Future Work

The first implementation is incorporating the FiLM conditioning mechanism to replace the current additive projection. Second, training with an explicit perturbation curriculum (Stage 4 of the project plan). Third, a deeper analysis of the hidden state geometry, specifically, whether the modulatory signal from the RA induces structured deformations of the activation manifold that correspond to distinct functional modes (growth, maintenance, regeneration).

4 Questions to answer in the future work

- Does the ring-attractor geometry, by constraining the controller state to a low-dimensional phase manifold, improve morphological stability and regeneration robustness compared to an unconstrained modulatory controller?
- Do the ring attractor state variables (a , b , d) develop spatially distinct

activation patterns that correlate with growth, maintenance, and regenerative phases, and does damage induce a measurable phase reset in (a,b)?

- Does the modulatory signal from the ring attractor bias the hidden representation of GenePropCA toward distinct functional modes, and can this be measured through the geometry of the hidden activation space?

5 Experimental setup

All the experiments were run on a google colab notebook with a NVIDIA T4 GPU.

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